Tribhuvan University

Faculty of Humanities & Social Sciences

OFFICE OF THE DEAN

Bachelor in Computer Applications

Course Title: **Numerical Methods**

Code No: Semester: **3**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: *Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.*

Group B

1. Derive the formula for integration using Simpson's 3/8 rule. Use Secant Method to estimate the root of equation $x^2 - 4x - 10 = 0$, with initial estimate $x_1 = 4$ and $x_2 = 2$.

2. What do you mean by boundary value problem? Use shooting method, solve the equation:$y'' = 6x^2$, with $y(0) = 1$ and $y(1) = 2$ in the interval $(0, 1)$ for $y(0.5)$ taking $h = 0.5$.

3. Write an algorithm and program to compute the interpolation using Lagrange Interpolation.

4. Show that the rate of convergence of Newton's Raphson method is quadratic.

5. The temperature of a metal strip was measured at various time intervals during heating and the values are given in the table below.Time('t' min)1234Temp('T' °C)7083100124If the relation between the time 't' and temperature 'T' is of the form: $T = be^{t/4} + a$. Estimate the temperature at $t = 6$ minute.

6. Given the following set of data points. Obtain the table of divided difference and use that table to estimate the value of $f(1.5)$.x12345$f(x) = x^3 - 1$072663124

7. Solve the following system of linear equation by Gauss Elimination with Pivoting.$2x + 2y + z = 6$$4x + 2y + 3z = 4$$x - y + 1 = 0$

8. Determine the Eigen Values and corresponding Eigen Vectors for the matrix.$$A = \begin{bmatrix} 1 & 6 & 1 \\ 1 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

9. The table below gives the values of distance travelled by a car at various time intervals during the initial running.Time('t' sec)56789Temp('T' °C)10.014.519.525.532.0Estimate the velocity and acceleration at time $t = 7$ sec.

10. Solve the following integral using trapezoidal rule for $n = 8$.$$I = \int_{2}^{4} (x^4 + 1)\, dx$$

11. Given the equation $y' = 3x^2 + 1$ with $y(1) = 2$, estimate $y(2)$ by Euler's Method using $h = 0.2$.

12. Solve the Poisson's Equation $\nabla^2 f = 2x^2y^2$ over the square domain $0 \leq x \leq 3$ and $0 \leq y \leq 3$ with $f = 0$ on the boundary and $h = 1$.